V Vinaykarthik

ma23btech11026@iith.ac.in | vinayvikram223@gmail.com | +91 8284891527 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.39
Mount Carmel School, Chandigarh	Class XII (CBSE)	2022	96.20%
Mount Carmel School, Chandigarh	Class X (CBSE)	2020	94.00%

SCHOLASTIC ACHIEVEMENTS

- Received the **Academic Excellence Award** from the director of IIT Hyderabad; ranked **1st in the institute** for two semesters and maintained a top 3 department rank throughout the undergraduate program
- Achieved a peak **Codeforces** rating of **1556** and solved **650+ problems**
- Achieved the **highest title, Champion of Champions**, at the International Abacus Competition in Singapore conducted by **SEMAS/GAJSMA**; secured **1st prize** in Thailand and Champion of Champions at the national level
- Secured **2nd state rank** in Punjab and **44th national rank** in the Wizard Spell Bee, Kolkata

WORK EXPERIENCE

SkanAI Labs Pvt. Ltd.

Data Science Intern

- Built a **Conversation Analyser** (FastAPI + PostgreSQL backend, React UI), a privacy-first platform for **C-suite leaders** with **executive insights** on communication cost, organizational health and social network analysis via a two-stage LLM pipeline.
- Quantified the **business value** at stake: **$31.8K in annualized communication waste (~53 FTE-days)** at the org level in a simulated 251-employee firm, 91% from re-sent attachments and unanswered threads, with **3 of 4 fixes needing little spend**.
- Applied **graph analytics** (Louvain, centrality, BFS) and **bottleneck analysis**, showing that nearly half of staff work in cross-department clusters and **under 20% are reached within a workday** despite 98% reachability.
- Surfaced **organizational health** risks by department: **1 in 8 sent emails covertly BCCs a manager** and importance flags lift read rates by **only 2.9 points**, pointing to low trust in open escalation and diluted urgency.
- Engineered **two hybrid LLM + rule-based data generators** (Outlook action logs; labeled emails in **6 languages**) across 5 industries, planting **ground-truth personas** to score analytics detectors against a known answer.
- Built an **LLM entity and intent extractor** (regex recall floor, recall-then-prune prompting, coverage re-ask) to recover dropped dates, numbers and IDs, backed by a **hand-labeled 216-entity gold set**

PROJECTS

Psychological Manipulation Detection for LLMs

- Explored and evaluated multiple approaches for detecting **conversational manipulation targeting LLMs** using the **4,000+ dialogue MentalManip dataset**, covering **11 fine-grained manipulation labels**
- Progressed from **LSTM/BiLSTM baselines (0.10-0.11 macro F1)** to feature-driven and similarity approaches, incorporating **rule-based Empath/VADER features**, sentence-transformer embeddings and **k-NN similarity search** to achieve **~0.17 macro-F1**
- Advanced to **Intent-Aware Prompting (IAP)** by separating surface dialogue from latent intent through **parallel LLM-based intent extraction**; achieved **0.857 F1, 83.3% recall and 16.7% FNR**, with a **30.5% reduction in FNR** over baselines

Bayesian Modelling for Trading Strategy Selection

- Modelled the success probabilities of various trading strategies using **Beta-Binomial Bayesian updating** across sequential intervals on **2,500 daily Airtel stock prices (2015-2025)**

- Extended the **Beta-Binomial model** to a **Normal-Normal conjugate formulation** incorporating return magnitude while maintaining success probabilities, enabling **posterior predictive analysis** of future trade outcomes
- Jointly estimated profitability across **5 trading strategies** using a **Dirichlet-Multinomial model**, followed by **10,000-draw Monte Carlo posterior-predictive simulation** to quantify future profitability uncertainty

Neural Network Approximation of Logistic Map Dynamics

- Investigated the **Universal Approximation Theorem** by implementing a 1-hidden-layer neural network in Python to study shallow neural-network approximation of mathematical functions
- Analyzed model behavior across **smooth and discontinuous functions** (step and extended-step), studying the effects of **domain interval, activation function,** and **hidden-layer capacity** on approximation accuracy
- Applied the model to **detect bifurcation points** in the **logistic map** and predict transitions into chaos, validating predictions against the known **bifurcation diagram**

SKILLS

- **Programming Languages:** C++, Python, SQL, C
- **Tools & Libraries:** NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Kaggle, Git, DBeaver, LaTeX

RELEVANT COURSES

- **Computer Science:** Approximation Algorithms, Algorithmic Techniques for Large Graphs, DBMS, OS-I, Computer Networks, Compilers-I, Data Structures, Algorithms, Theory of Computation
- **Mathematics:** Probability Theory, Bayesian Statistics, Linear Algebra, Predictive Modeling, Convex Optimization, Vector Calculus, Differential Equations, Combinatorics, Number Theory
- **Miscellaneous:** Foundations of Machine Learning, Natural Language Processing, Fuzzy Logic